1). Install Terraform on your PC

Implementation steps

1. Open the official Terraform installation page.
2. Download the Terraform ZIP file for Windows AMD64.
3. Extract the downloaded ZIP file.
4. Create the folder C:\terraform.
5. Copy terraform.exe into C:\terraform.
6. Open Windows Search and select Edit the system environment variables.
7. Click Environment Variables.
8. Under User variables, select Path and click Edit.
9. Click New, add C:\terraform, and save the changes by clicking OK on all open windows.
10. Open a new Git Bash terminal.
11. Run:

terraform -version

12. Verify that Terraform displays its version and operating system.

verification output:

Terraform v1.17.0

on windows_amd64



2). Execute all the templates shown in video.

A. Create the Terraform project

1. Open Git Bash.
2. Create the project directory and navigate into it:

```
mkdir C:/terraform-basic
cd C:/terraform-basic
```

3. Open the folder in Visual Studio Code.
4. Create a file named main.tf.
5. Add the following configuration and save it:

```
resource "local_file" "my_pet" {
 filename = "pets.txt"
 content  = "I love pets"
}
```

B. Initialize Terraform

Run:

```
terraform init
```

Expected result: Terraform initializes the working directory and downloads the required provider plugin, hashicorp/local.

C. Preview the configuration

Run:

```
terraform plan
```

Verify that Terraform proposes creating one resource:

Plan: 1 to add, 0 to change, 0 to destroy.

D. Apply the configuration

Run:

```
terraform apply
```

1. Review the proposed changes.
2. Type yes when prompted.
3. Verify that the file was created:

```
ls
```

cat pets.txt

Expected file content:

I love pets

E. Modify the existing configuration

1. Open main.tf.
2. Change the content to "I hate pets":

```
resource "local_file" "my_pet" {
 filename = "pets.txt"
 content  = "I hate pets"
}
```

3. Save the file.
4. Preview the change:

terraform plan

5. Apply the change:

terraform apply

6. Type yes when prompted.
7. Verify the updated file:

cat pets.txt

Expected output:

I hate pets

F. Work with the Random Pet provider

1. Add this resource to main.tf, keeping the existing local_file resource:

```
resource "random_pet" "my_pet" {
 prefix    = "Mr"
 separator = "."
 length    = 1
```

}

2. Save the file.
3. Initialize the new provider:

terraform init

4. Format the configuration:

terraform fmt

5. Preview the changes:

terraform plan

6. Apply the configuration:

terraform apply

7. Type yes when prompted.

Terraform generates a random pet name with the prefix Mr.

G. Inspect Terraform state

List the managed resources:

terraform state list

Expected resource addresses:

local_file.my_pet

random_pet.my_pet

Inspect the random pet resource:

terraform state show random_pet.my_pet

Your output included:

```
resource "random_pet" "my_pet" {
  id        = "Mr.toucan"
  length    = 1
  prefix    = "Mr"
  separator = "."
}
```

H. Destroy the managed resources

Run:

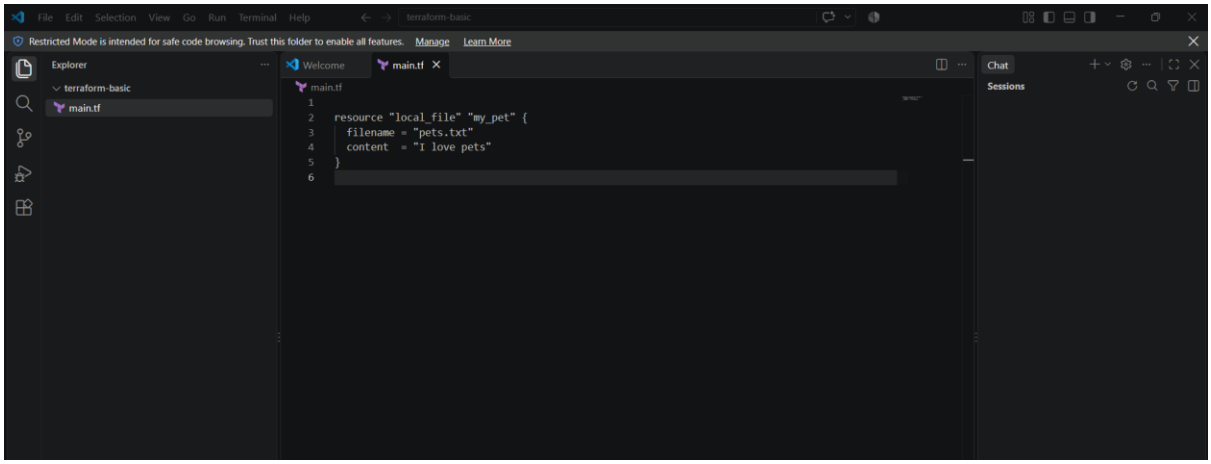
terraform destroy

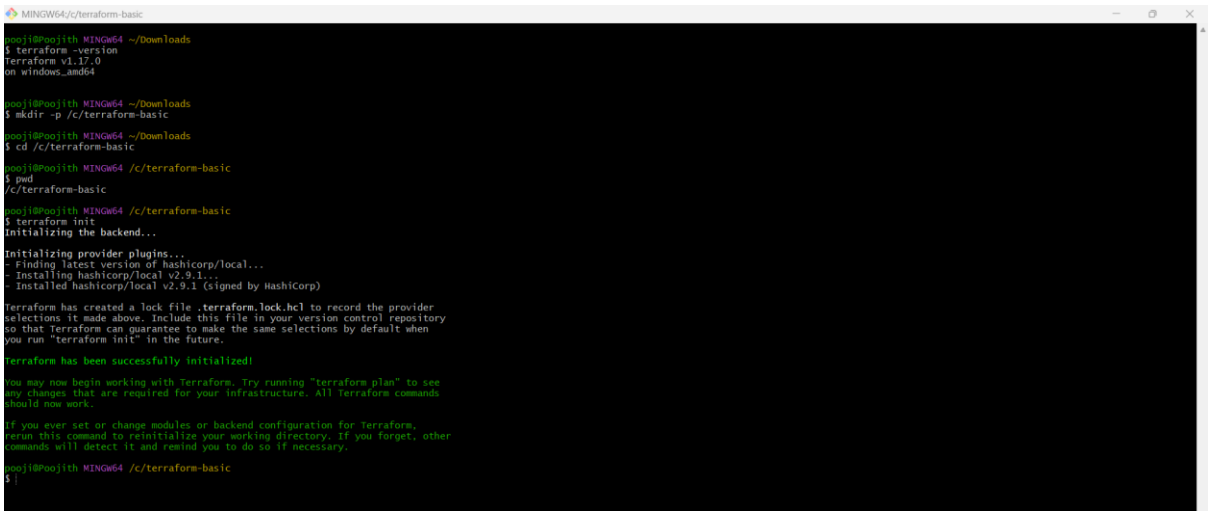
1. Review the resources Terraform proposes to remove.
2. Type yes when prompted.
3. Verify the state is empty:

terraform state list

final output was:

Destroy complete! Resources: 2 destroyed.





```
pooji@Poojith MINGW64 /c/terraform-basic
$ terraform plan

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # local_file.my_pet will be created
  + resource "local_file" "my_pet" {
      + content              = "I love pets"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "pets.txt"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

─────────────────────────────────────────────────────────────────────────────

Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.

pooji@Poojith MINGW64 /c/terraform-basic
$
```

```
pooji@Poojith MINGW64 /c/terraform-basic
$ terraform apply

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # local_file.my_pet will be created
  + resource "local_file" "my_pet" {
      + content              = "I love pets"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "pets.txt"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.my_pet: Creating...
local_file.my_pet: Creation complete after 0s [id=4246687863ee17bf4daf5fc376ab01222e989aca]

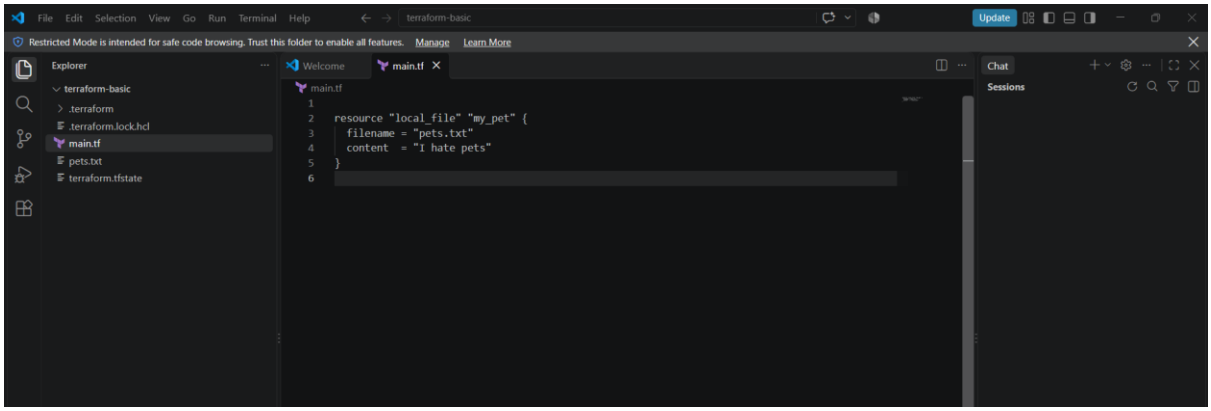
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

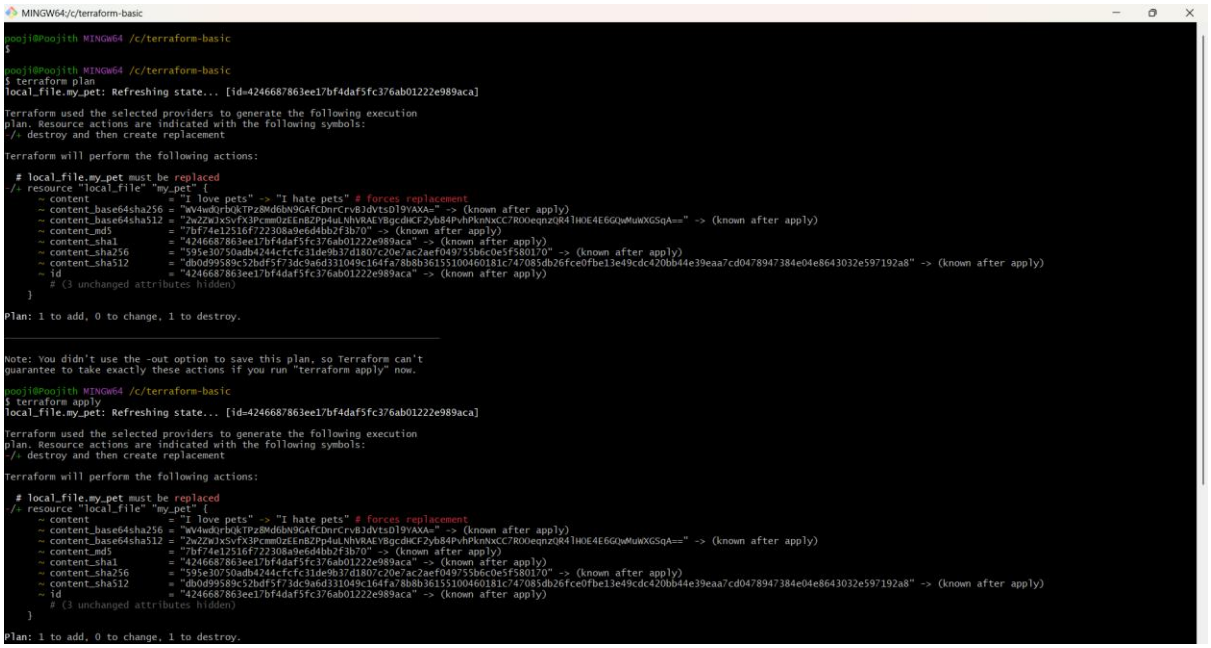
pooji@Poojith MINGW64 /c/terraform-basic
$
```

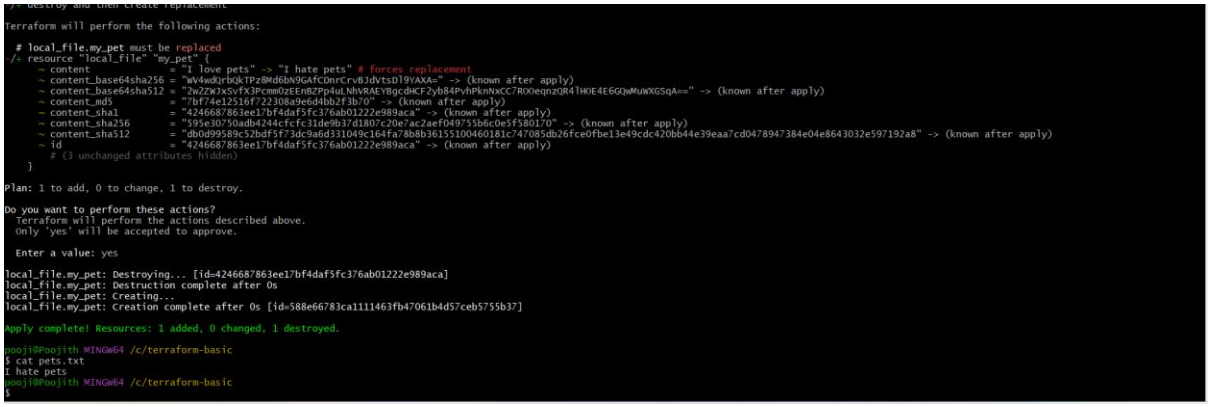
```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

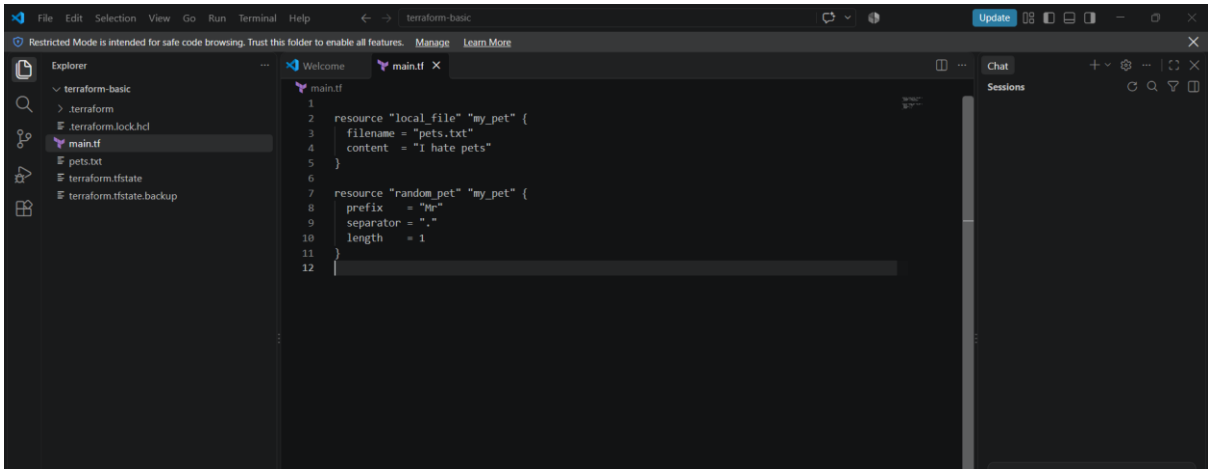
pooji@Poojith MINGW64 /c/terraform-basic
$ cat pets.txt
I love pets
pooji@Poojith MINGW64 /c/terraform-basic
$ ls -l pets.txt
-rw-r--r-- 1 pooji 197609 11 Oct 26 17:35 pets.txt

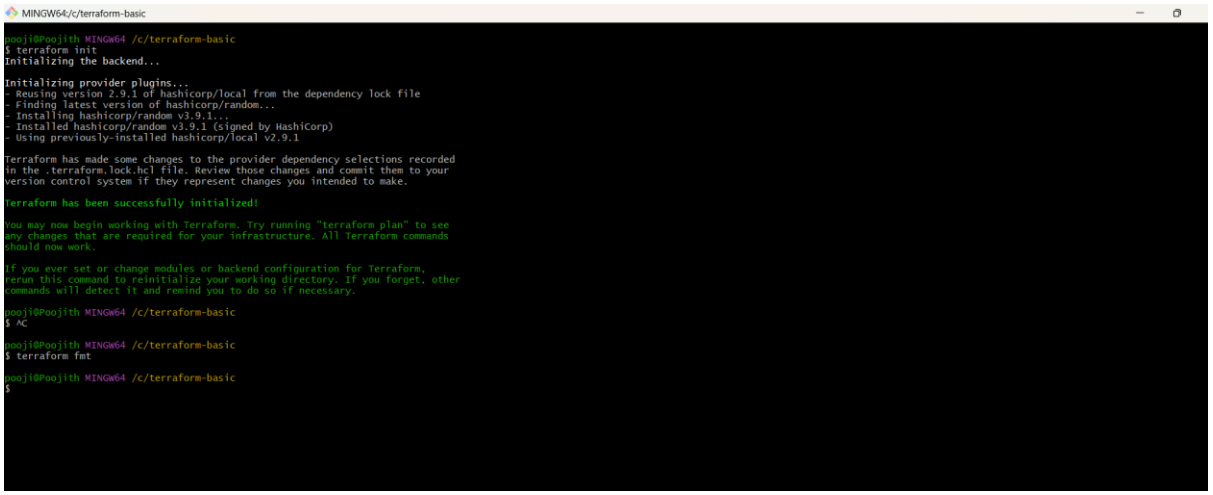
pooji@Poojith MINGW64 /c/terraform-basic
$
```

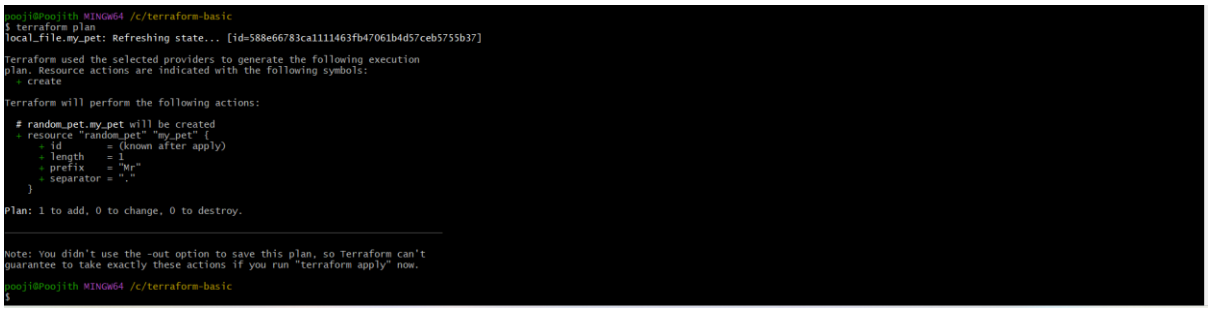


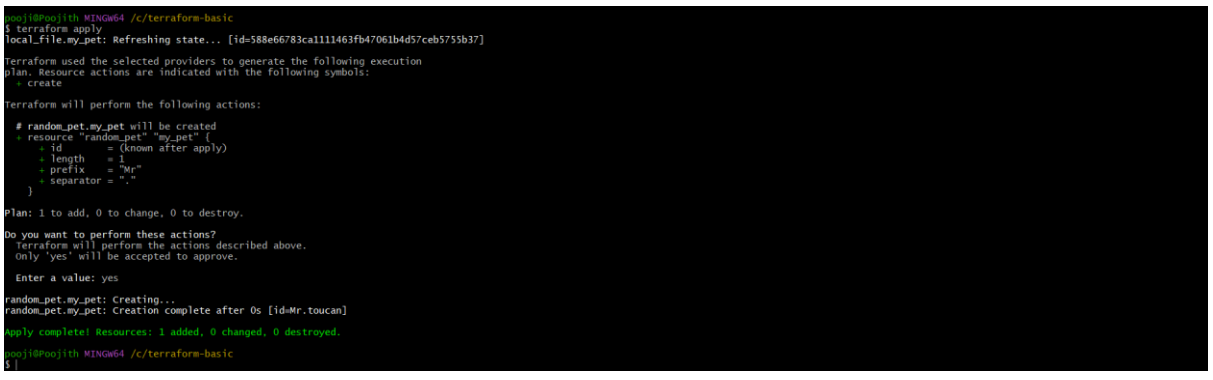

MINGW64:/c/terraform-basic
pooji@Poojith MINGW64 /c/terraform-basic
$
pooji@Poojith MINGW64 /c/terraform-basic
$ terraform plan
local_file.my_pet: Refreshing state... [id=4246687863ee17bf4daf5fc376ab01222e989aca]
Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement
Terraform will perform the following actions:
local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
~ content = "I love pets" -> "I hate pets" # forces replacement
~ content_base64sha256 = "wV4wdQrbQkTPz8Md6bN9GAfCDnrCrvBJdVtsDl9YAXA=" -> (known after apply)
~ content_base64sha512 = "2w2ZWJxSvfX3Pcmm0zEEnBZPp4uLNhVRAEYBgcdHCF2yb84PvhPknNxCC7ROOeqnzQR4lHOE4E6GQwMuWXGSqA==" -> (known after apply)
~ content_md5 = "7bf74e12516f722308a9e6d4bb2f3b70" -> (known after apply)
~ content_sha1 = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
~ content_sha256 = "595e30750adb4244cfcfc31de9b37d1807c20e7ac2aef049755b6c0e5f580170" -> (known after apply)
~ content_sha512 = "db0d99589c52bdf5f73dc9a6d331049c164fa78b8b36155100460181c747085db26fce0fbe13e49cdc420bb44e39eaa7cd0478947384e04e8643032e597192a8" -> (known after apply)
~ id = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
(3 unchanged attributes hidden)
}
Plan: 1 to add, 0 to change, 1 to destroy.
Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
pooji@Poojith MINGW64 /c/terraform-basic
$ terraform apply
local_file.my_pet: Refreshing state... [id=4246687863ee17bf4daf5fc376ab01222e989aca]
Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement
Terraform will perform the following actions:
local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
~ content = "I love pets" -> "I hate pets" # forces replacement
~ content_base64sha256 = "wV4wdQrbQkTPz8Md6bN9GAfCDnrCrvBJdVtsDl9YAXA=" -> (known after apply)
~ content_base64sha512 = "2w2ZWJxSvfX3Pcmm0zEEnBZPp4uLNhVRAEYBgcdHCF2yb84PvhPknNxCC7ROOeqnzQR4lHOE4E6GQwMuWXGSqA==" -> (known after apply)
~ content_md5 = "7bf74e12516f722308a9e6d4bb2f3b70" -> (known after apply)
~ content_sha1 = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
~ content_sha256 = "595e30750adb4244cfcfc31de9b37d1807c20e7ac2aef049755b6c0e5f580170" -> (known after apply)
~ content_sha512 = "db0d99589c52bdf5f73dc9a6d331049c164fa78b8b36155100460181c747085db26fce0fbe13e49cdc420bb44e39eaa7cd0478947384e04e8643032e597192a8" -> (known after apply)
~ id = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
(3 unchanged attributes hidden)
}
Plan: 1 to add, 0 to change, 1 to destroy.

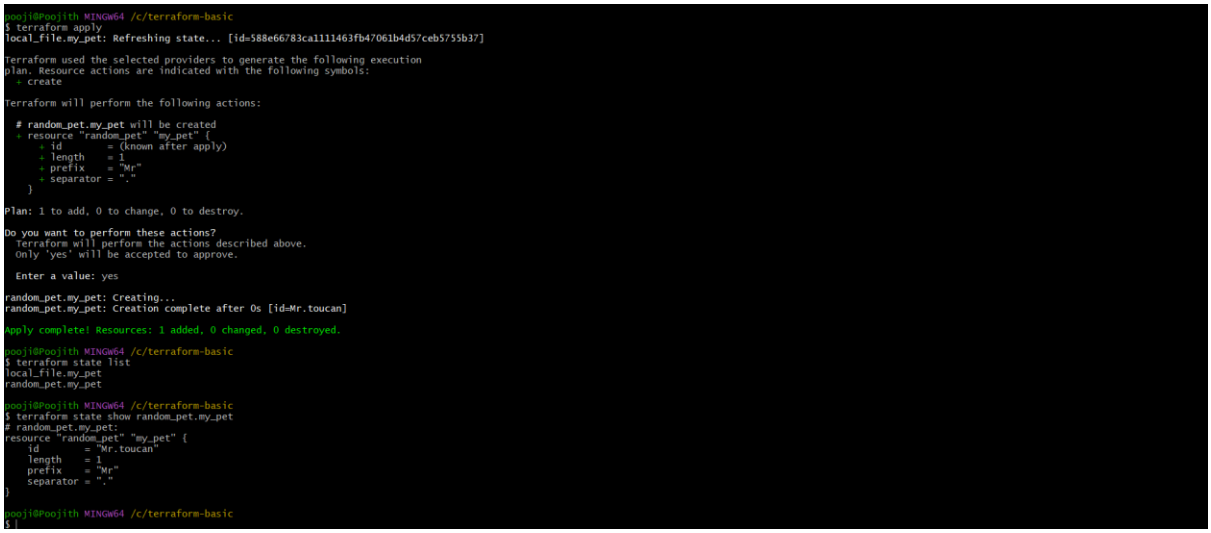

-/+ destroy and then create replacement
Terraform will perform the following actions:
local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
~ content = "I love pets" -> "I hate pets" # forces replacement
~ content_base64sha256 = "wV4wdQrbQkTPz8Md6bN9GAfCDnrCrvBJdVtsDl9YAXA=" -> (known after apply)
~ content_base64sha512 = "2w2ZWJxSvfX3Pcmm0zEEnBZPp4uLNhVRAEYBgcdHCF2yb84PvhPknNxCC7ROOeqnzQR4lHOE4E6GQwMuWXGSqA==" -> (known after apply)
~ content_md5 = "7bf74e12516f722308a9e6d4bb2f3b70" -> (known after apply)
~ content_sha1 = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
~ content_sha256 = "595e30750adb4244cfcfc31de9b37d1807c20e7ac2aef049755b6c0e5f580170" -> (known after apply)
~ content_sha512 = "db0d99589c52bdf5f73dc9a6d331049c164fa78b8b36155100460181c747085db26fce0fbe13e49cdc420bb44e39eaa7cd0478947384e04e8643032e597192a8" -> (known after apply)
~ id = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
(3 unchanged attributes hidden)
}
Plan: 1 to add, 0 to change, 1 to destroy.
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
local_file.my_pet: Destroying... [id=4246687863ee17bf4daf5fc376ab01222e989aca]
local_file.my_pet: Destruction complete after 0s
local_file.my_pet: Creating...
local_file.my_pet: Creation complete after 0s [id=588e66783ca1111463fb47061b4d57ceb5755b37]
Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
pooji@Poojith MINGW64 /c/terraform-basic
$ cat pets.txt
I hate pets
pooji@Poojith MINGW64 /c/terraform-basic
$


File Edit Selection View Go Run Terminal Help
terraform-basic
Update
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
Explorer
terraform-basic
.terraform
.terraform.lock.hcl
main.tf
pets.txt
terraform.tfstate
terraform.tfstate.backup
Welcome
main.tf
main.tf
resource "local_file" "my_pet" {
filename = "pets.txt"
content = "I hate pets"
}
resource "random_pet" "my_pet" {
prefix = "Mr"
separator = "."
length = 1
}
Chat
Sessions


MINGW64:/c/terraform-basic
pooji@Poojith MINGW64 /c/terraform-basic
$ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing version 2.9.1 of hashicorp/local from the dependency lock file
- Finding latest version of hashicorp/random...
- Installing hashicorp/random v3.9.1...
- Installed hashicorp/random v3.9.1 (signed by HashiCorp)
- Using previously-installed hashicorp/local v2.9.1
Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.
Terraform has been successfully initialized!
You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
pooji@Poojith MINGW64 /c/terraform-basic
$ AC
pooji@Poojith MINGW64 /c/terraform-basic
$ terraform fmt
pooji@Poojith MINGW64 /c/terraform-basic
$


pooji@Poojith MINGW64 /c/terraform-basic
$ terraform plan
local_file.my_pet: Refreshing state... [id=588e66783ca1111463fb47061b4d57ceb5755b37]
Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
random_pet.my_pet will be created
+ resource "random_pet" "my_pet" {
+ id = (known after apply)
+ length = 1
+ prefix = "Mr"
+ separator = "."
}
Plan: 1 to add, 0 to change, 0 to destroy.
Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
pooji@Poojith MINGW64 /c/terraform-basic
$


pooji@Poojith MINGW64 /c/terraform-basic
$ terraform apply
local_file.my_pet: Refreshing state... [id=588e66783ca1111463fb47061b4d57ceb5755b37]
Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
random_pet.my_pet will be created
+ resource "random_pet" "my_pet" {
+ id = (known after apply)
+ length = 1
+ prefix = "Mr"
+ separator = "."
}
Plan: 1 to add, 0 to change, 0 to destroy.
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
random_pet.my_pet: Creating...
random_pet.my_pet: Creation complete after 0s [id=Mr.toucan]
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
pooji@Poojith MINGW64 /c/terraform-basic
$





3). Note down below points, Terraform Init Terraform Plan Terraform Apply Terraform Provider.

1. Terraform Init

Purpose: Initializes the Terraform working directory.

- Prepares the directory for Terraform operations.
- Downloads the required provider plugins.
- Creates or updates .terraform.lock.hcl to record provider selections.

Command:

terraform init

Expected result: Terraform has been successfully initialized!

2. Terraform Plan

Purpose: Previews the changes Terraform intends to make before applying them.

- Reads the configuration and checks the current state.
- Shows which resources Terraform plans to create, change, replace, or destroy.
- Does not apply the proposed changes.

Command:

terraform plan

Example output:

Plan: 1 to add, 0 to change, 0 to destroy.

3. Terraform Apply

Purpose: Applies the planned changes and creates, updates, or deletes resources.

- Displays the proposed changes.
- Normally asks for confirmation before making changes.
- Updates the Terraform state after successful operations.

Command:

terraform apply

Type yes when prompted.

Example output:

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

4. Terraform Provider

Purpose: A provider is a plugin that enables Terraform to communicate with and manage resources through a particular platform or service.

Examples from your trainee's document:

Provider	Purpose
local	Manages local files

random	Generates random values
aws	Manages AWS resources
azurerm	Manages Azure resources
google	Manages Google Cloud resources

In your practical, Terraform installed hashicorp/local v2.9.1 and hashicorp/random v3.9.1.

4). Integrate a sample Terraform template in Jenkins.

A. Launch and configure EC2

1. Sign in to the AWS Management Console.

2. Open EC2 → Instances → Launch instance.

3. Configure the instance:

Setting	Value used
Name	Terraform-Jenkins-Lab
Operating system	Amazon Linux 2023
Instance type	t3.small
Region	US East (N. Virginia), us-east-1
Login	ec2-user

4. Select or create a key pair if required.

5. Configure a security group with these inbound rules:

Type	Port	Source
SSH	22	My IP
Custom TCP	8080	My IP

6. Launch the instance and wait until its state is Running.

7. Connect through EC2 Instance Connect.

B. Install Java and Jenkins

1. Verify the operating system and login user:

cat /etc/os-release

whoami

2. Update the system packages:

sudo dnf update -y

3. Install Java 21:

sudo dnf install -y java-21-amazon-corretto fontconfig

4. Verify Java:

java -version

Your installed version was Amazon Corretto 21.0.13.

5. Add the Jenkins repository:

sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo

6. Import the repository signing key:

sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2026.key

7. Install Jenkins:

sudo dnf install jenkins -y

8. Enable and start Jenkins:

sudo systemctl enable --now jenkins

9. Verify the service:

sudo systemctl status jenkins

Confirm that the output contains:

Active: active (running)

10. Retrieve the initial administrator password:

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

11. Open Jenkins in a browser:

http://<EC2-PUBLIC-IP>:8080

12. Enter the initial password, install suggested plugins, create an administrator account, and complete the setup wizard.

C. Install Terraform on EC2

1. Install the repository management utility:

sudo dnf install -y dnf-plugins-core

2. Add the HashiCorp repository:

sudo dnf config-manager --add-repo
https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo

3. Install Terraform:

sudo dnf install -y terraform

4. Verify the installation:

terraform -version

5. Check the Terraform executable path:

which terraform

6. Verify that the Jenkins user can access Terraform:

sudo -u jenkins terraform -version

D. Create the Jenkins Pipeline job

1. Open the Jenkins dashboard.
2. Click New Item.
3. Enter the job name:

terraform-demo

4. Select Pipeline.
5. Click OK.
6. Scroll to the Pipeline section.
7. Set Definition to Pipeline script.

8. Paste the following script. This is the working version you executed successfully:

```
pipeline {
  agent any

  stages {
    stage('Check Terraform') {
      steps {
        sh 'terraform -version'
      }
    }

    stage('Create Terraform Template') {
      steps {
        writeFile file: 'main.tf', text: '''
resource "local_file" "jenkins_output" {
 filename = "jenkins-output.txt"
 content  = "Terraform executed successfully through Jenkins"
}
'''
      }
    }

    stage('Terraform Init') {
      steps {
        sh 'terraform init -input=false'
      }
    }
```

```
        stage('Terraform Validate') {
            steps {
                sh 'terraform fmt -check'
                sh 'terraform validate'
            }
        }

        stage('Terraform Plan') {
            steps {
                sh 'terraform plan -input=false -out=tfplan'
            }
        }

        stage('Terraform Apply') {
            steps {
                sh 'terraform apply -input=false -auto-approve tfplan'
            }
        }

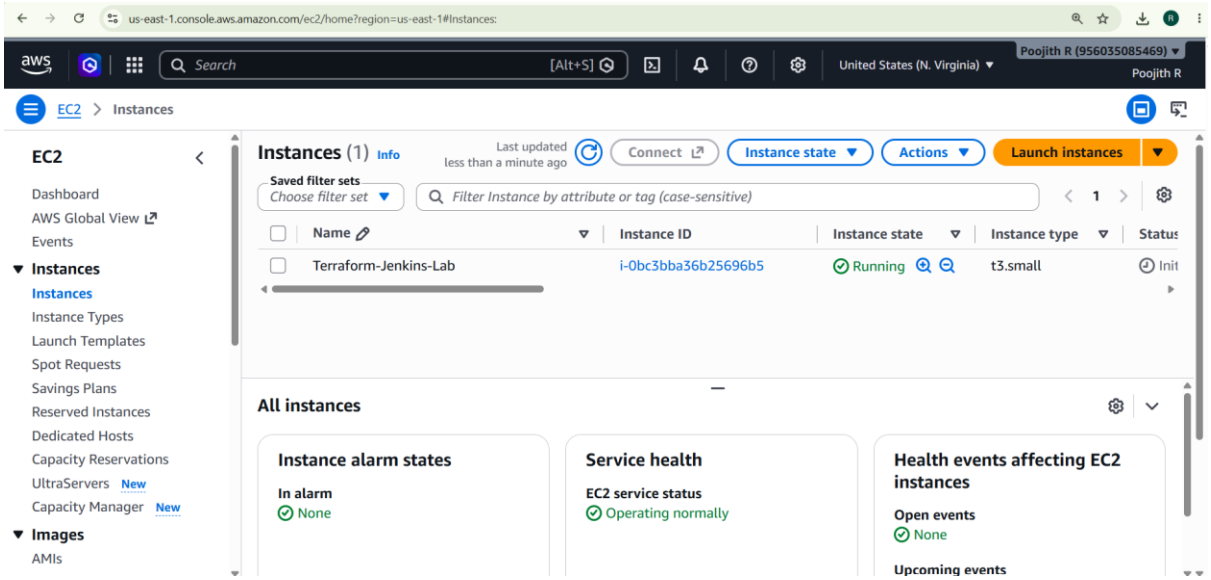
        stage('Verify Output') {
            steps {
                sh 'cat jenkins-output.txt'
            }
        }
    }
}
```

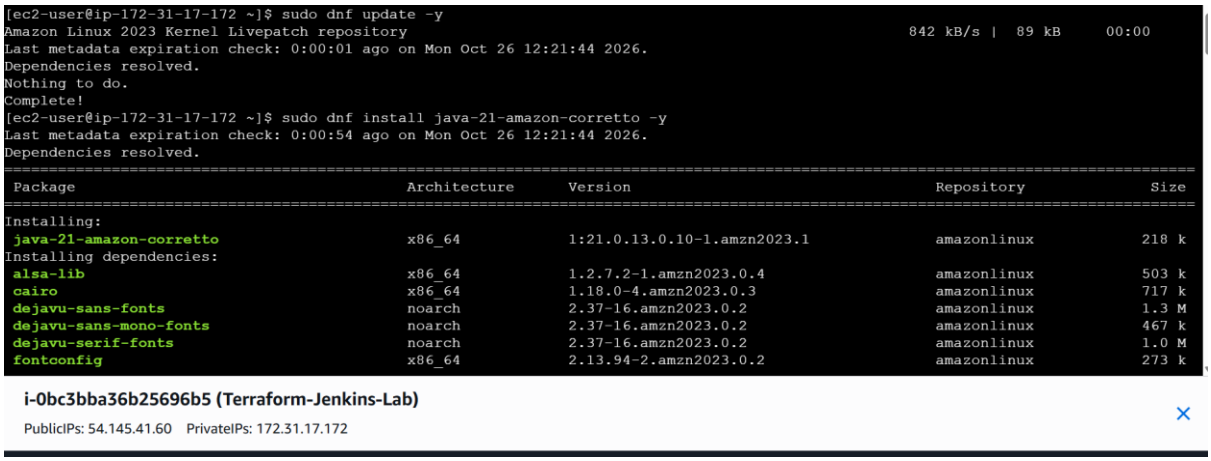
9. Click Save.

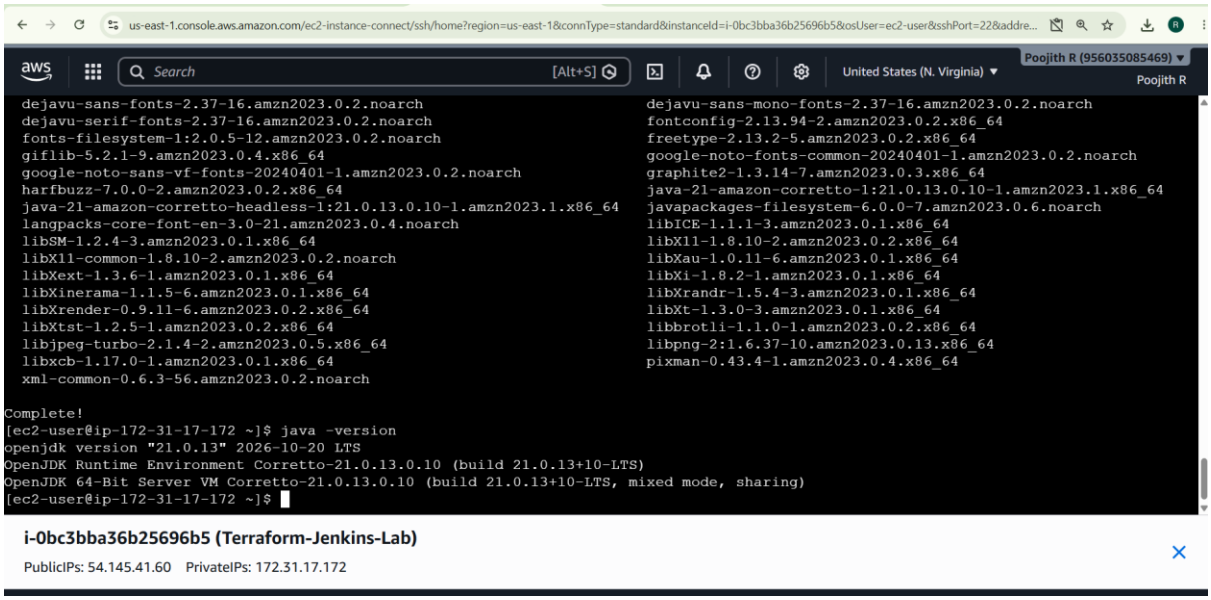
E. Execute and verify the Jenkins job

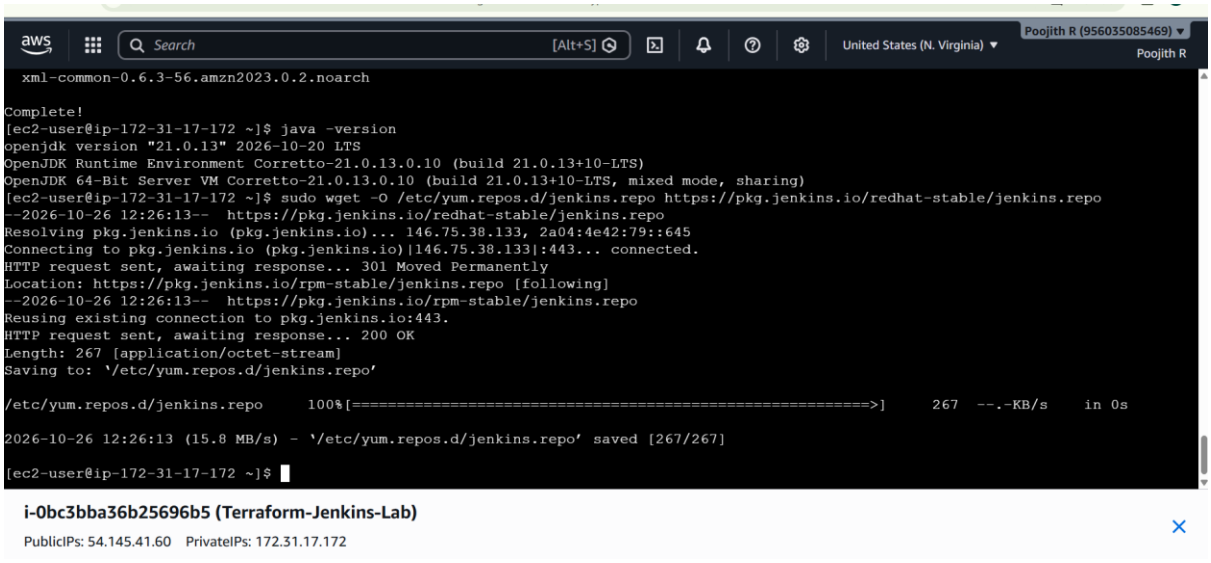
1. Open the terraform-demo job.
2. Click Build Now.
3. Wait for the build to finish.
4. Click the latest build number.
5. Open Console Output.
6. Verify that Terraform initializes, validates, plans, and applies the configuration.

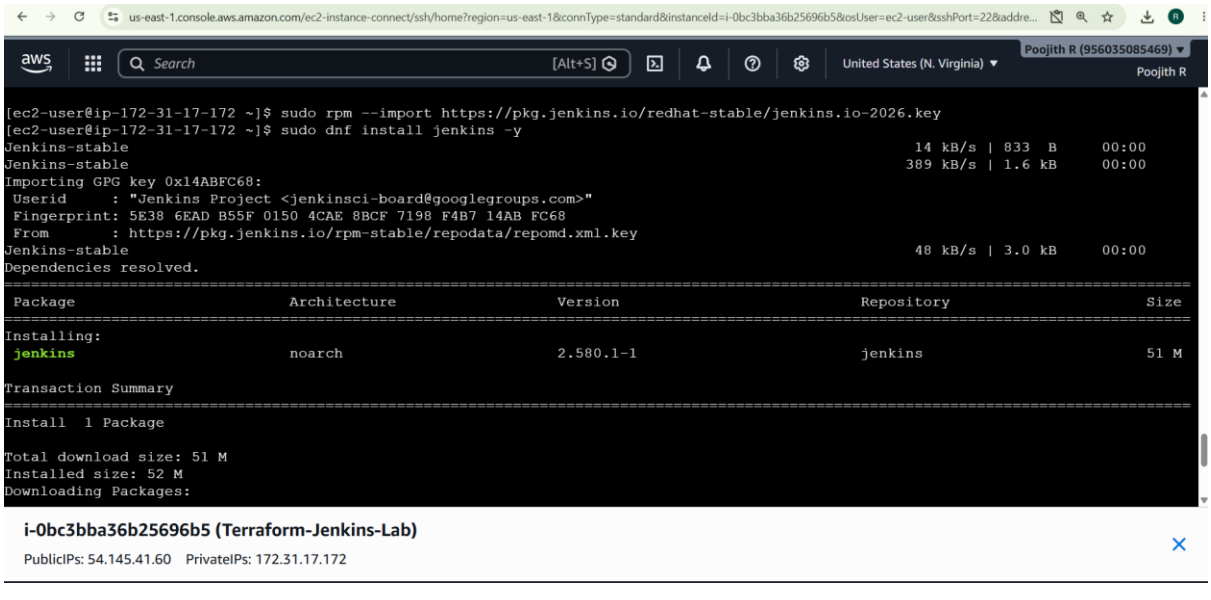
Result: Terraform was integrated with Jenkins on EC2, and a sample Terraform template ran through a Jenkins Pipeline to create a local output file in the Jenkins workspace.

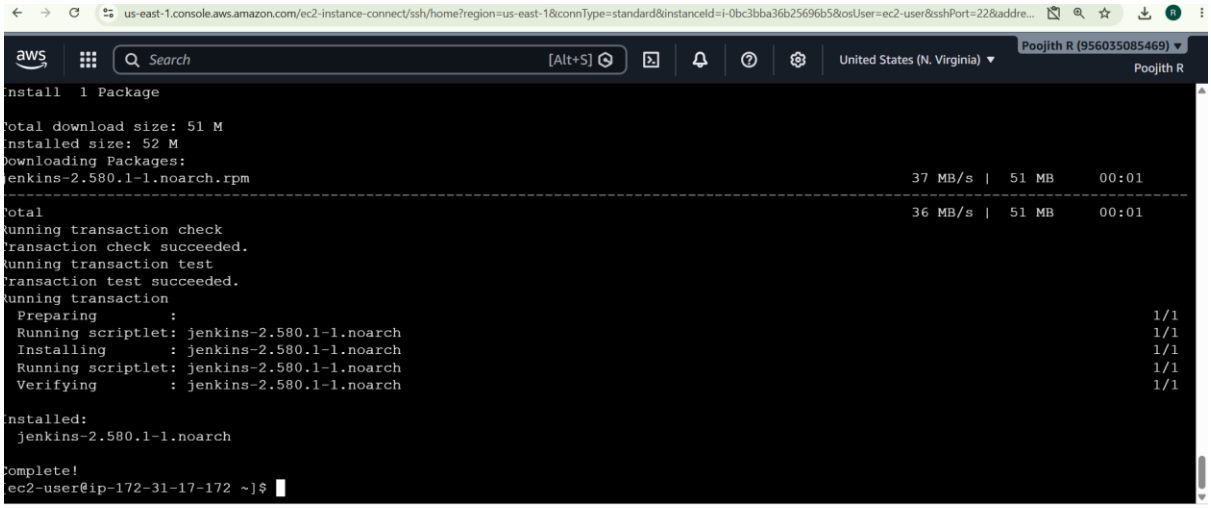


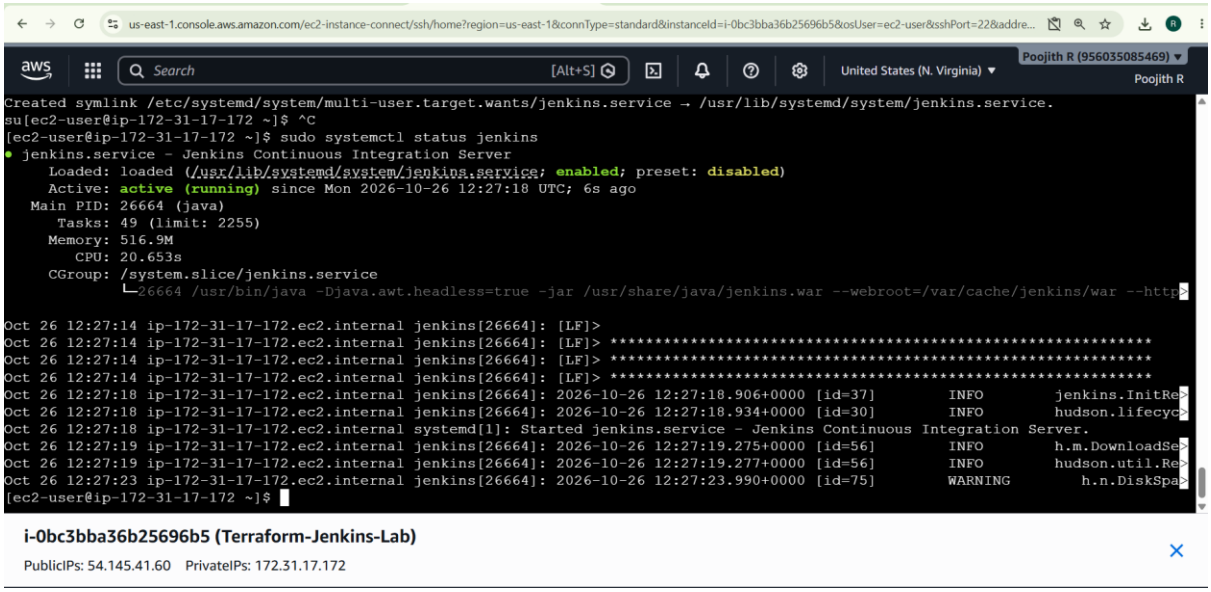


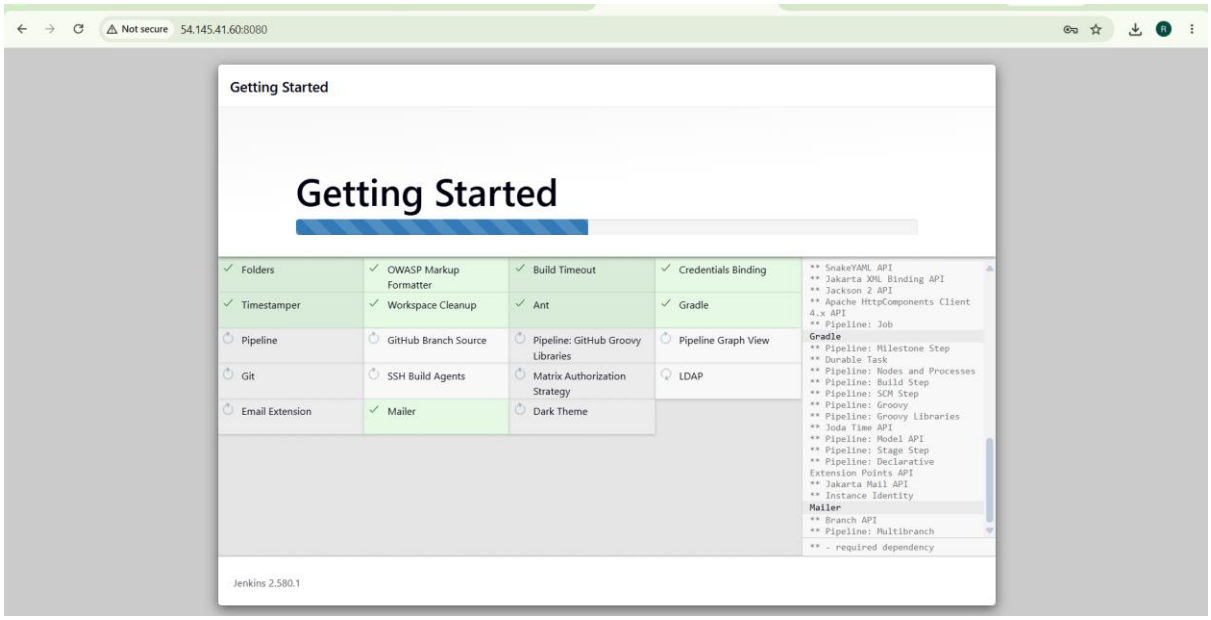

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0bc3bba36b25696b5&osUser=ec2-user&sshPort=22&addre...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
dejavu-sans-fonts-2.37-16.amzn2023.0.2.noarch
dejavu-sans-mono-fonts-2.37-16.amzn2023.0.2.noarch
dejavu-serif-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
fonts-filesystem-1:2.0.5-12.amzn2023.0.2.noarch
freetype-2.13.2-5.amzn2023.0.2.x86_64
giflib-5.2.1-9.amzn2023.0.4.x86_64
google-noto-fonts-common-20240401-1.amzn2023.0.2.noarch
google-noto-sans-vf-fonts-20240401-1.amzn2023.0.2.noarch
graphite2-1.3.14-7.amzn2023.0.3.x86_64
harfbuzz-7.0.0-2.amzn2023.0.2.x86_64
java-21-amazon-corretto-1:21.0.13.0.10-1.amzn2023.1.x86_64
java-21-amazon-corretto-headless-1:21.0.13.0.10-1.amzn2023.1.x86_64
javapackages-filesystem-6.0.0-7.amzn2023.0.6.noarch
langpacks-core-font-en-3.0-21.amzn2023.0.4.noarch
libICE-1.1.1-3.amzn2023.0.1.x86_64
libSM-1.2.4-3.amzn2023.0.1.x86_64
libX11-1.8.10-2.amzn2023.0.2.x86_64
libX11-common-1.8.10-2.amzn2023.0.2.noarch
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXi-1.8.2-1.amzn2023.0.1.x86_64
libXinerama-1.1.5-6.amzn2023.0.1.x86_64
libXrandr-1.5.4-3.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.2.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libXtst-1.2.5-1.amzn2023.0.2.x86_64
libbrotli-1.1.0-1.amzn2023.0.2.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
libpng-2:1.6.37-10.amzn2023.0.13.x86_64
libxcb-1.17.0-1.amzn2023.0.1.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
xml-common-0.6.3-56.amzn2023.0.2.noarch
Complete!
[ec2-user@ip-172-31-17-172 ~]$ java -version
openjdk version "21.0.13" 2026-10-20 LTS
OpenJDK Runtime Environment Corretto-21.0.13.0.10 (build 21.0.13+10-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.13.0.10 (build 21.0.13+10-LTS, mixed mode, sharing)
[ec2-user@ip-172-31-17-172 ~]$
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172

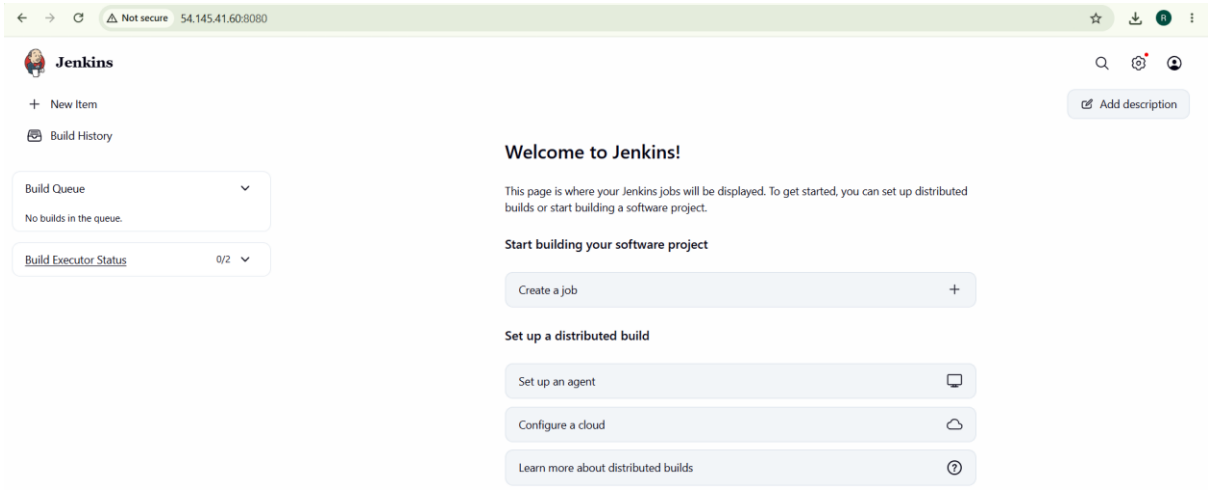

aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
xml-common-0.6.3-56.amzn2023.0.2.noarch
Complete!
[ec2-user@ip-172-31-17-172 ~]$ java -version
openjdk version "21.0.13" 2026-10-20 LTS
OpenJDK Runtime Environment Corretto-21.0.13.0.10 (build 21.0.13+10-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.13.0.10 (build 21.0.13+10-LTS, mixed mode, sharing)
[ec2-user@ip-172-31-17-172 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo
--2026-10-26 12:26:13-- https://pkg.jenkins.io/redhat-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.38.133|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://pkg.jenkins.io/rpm-stable/jenkins.repo [following]
--2026-10-26 12:26:13-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Reusing existing connection to pkg.jenkins.io:443.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'
/etc/yum.repos.d/jenkins.repo 100%[=>] 267 --.-KB/s in 0s
2026-10-26 12:26:13 (15.8 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]
[ec2-user@ip-172-31-17-172 ~]$
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0bc3bba36b25696b5&osUser=ec2-user&sshPort=22&addre...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
[ec2-user@ip-172-31-17-172 ~]$ sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2026.key
[ec2-user@ip-172-31-17-172 ~]$ sudo dnf install jenkins -y
Jenkins-stable 14 kB/s | 833 B 00:00
Jenkins-stable 389 kB/s | 1.6 kB 00:00
Importing GPG key 0x14ABFC68:
Userid : "Jenkins Project <jenkinsci-board@googlegroups.com>"
Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
From : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Jenkins-stable 48 kB/s | 3.0 kB 00:00
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
jenkins noarch 2.580.1-1 jenkins 51 M
Transaction Summary
Install 1 Package
Total download size: 51 M
Installed size: 52 M
Downloading Packages:
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0bc3bba36b25696b5&osUser=ec2-user&sshPort=22&addre...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
nstall 1 Package
otal download size: 51 M
nstalled size: 52 M
ownloading Packages:
enkins-2.580.1-1.noarch.rpm 37 MB/s | 51 MB 00:01
otal 36 MB/s | 51 MB 00:01
unning transaction check
ransaction check succeeded.
unning transaction test
ransaction test succeeded.
unning transaction
Preparing :
Running scriptlet: jenkins-2.580.1-1.noarch
Installing : jenkins-2.580.1-1.noarch
Running scriptlet: jenkins-2.580.1-1.noarch
Verifying : jenkins-2.580.1-1.noarch
nstalled:
jenkins-2.580.1-1.noarch
omplete!
ec2-user@ip-172-31-17-172 ~]$
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172

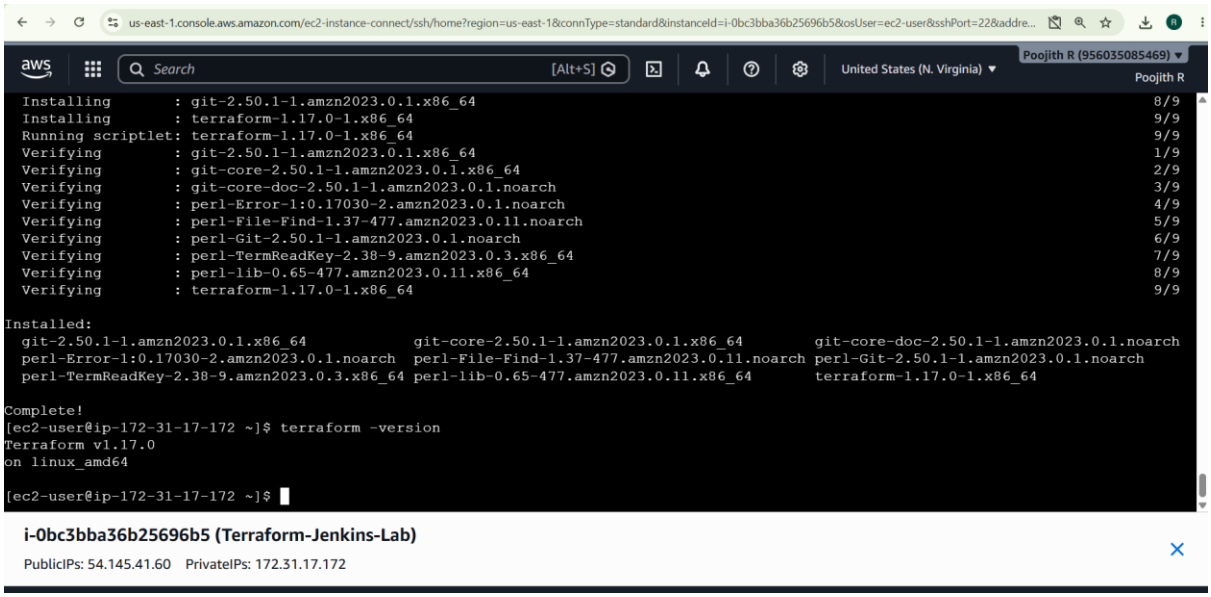

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0bc3bba36b25696b5&osUser=ec2-user&sshPort=22&addre...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
su[ec2-user@ip-172-31-17-172 ~]$ ^C
[ec2-user@ip-172-31-17-172 ~]$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
Active: active (running) since Mon 2026-10-26 12:27:18 UTC; 6s ago
Main PID: 26664 (java)
Tasks: 49 (limit: 2255)
Memory: 516.9M
CPU: 20.653s
CGroup: /system.slice/jenkins.service
└─26664 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --http
Oct 26 12:27:14 ip-172-31-17-172.ec2.internal jenkins[26664]: [LF]>
Oct 26 12:27:14 ip-172-31-17-172.ec2.internal jenkins[26664]: [LF]> **
Oct 26 12:27:14 ip-172-31-17-172.ec2.internal jenkins[26664]: [LF]> **
Oct 26 12:27:14 ip-172-31-17-172.ec2.internal jenkins[26664]: [LF]> **
Oct 26 12:27:18 ip-172-31-17-172.ec2.internal jenkins[26664]: 2026-10-26 12:27:18.906+0000 [id=37] INFO jenkins.InitRe
Oct 26 12:27:18 ip-172-31-17-172.ec2.internal jenkins[26664]: 2026-10-26 12:27:18.934+0000 [id=30] INFO hudson.lifecyc
Oct 26 12:27:18 ip-172-31-17-172.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 26 12:27:19 ip-172-31-17-172.ec2.internal jenkins[26664]: 2026-10-26 12:27:19.275+0000 [id=56] INFO h.m.DownloadSe
Oct 26 12:27:19 ip-172-31-17-172.ec2.internal jenkins[26664]: 2026-10-26 12:27:19.277+0000 [id=56] INFO hudson.util.Re
Oct 26 12:27:23 ip-172-31-17-172.ec2.internal jenkins[26664]: 2026-10-26 12:27:23.990+0000 [id=75] WARNING h.n.DiskSpa
[ec2-user@ip-172-31-17-172 ~]$
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172

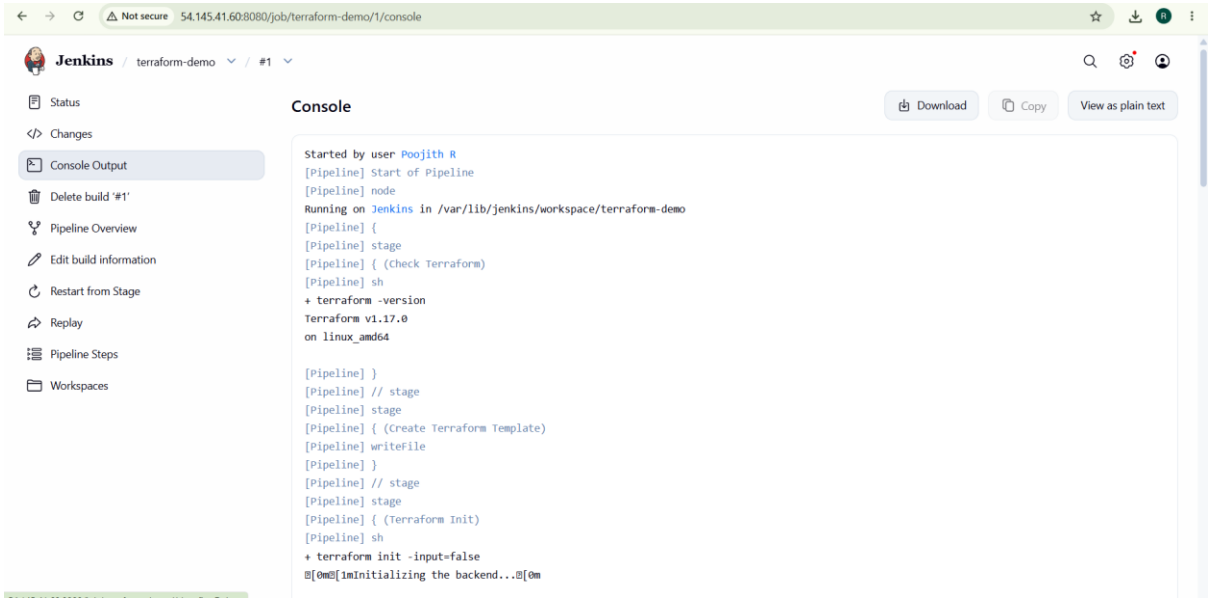


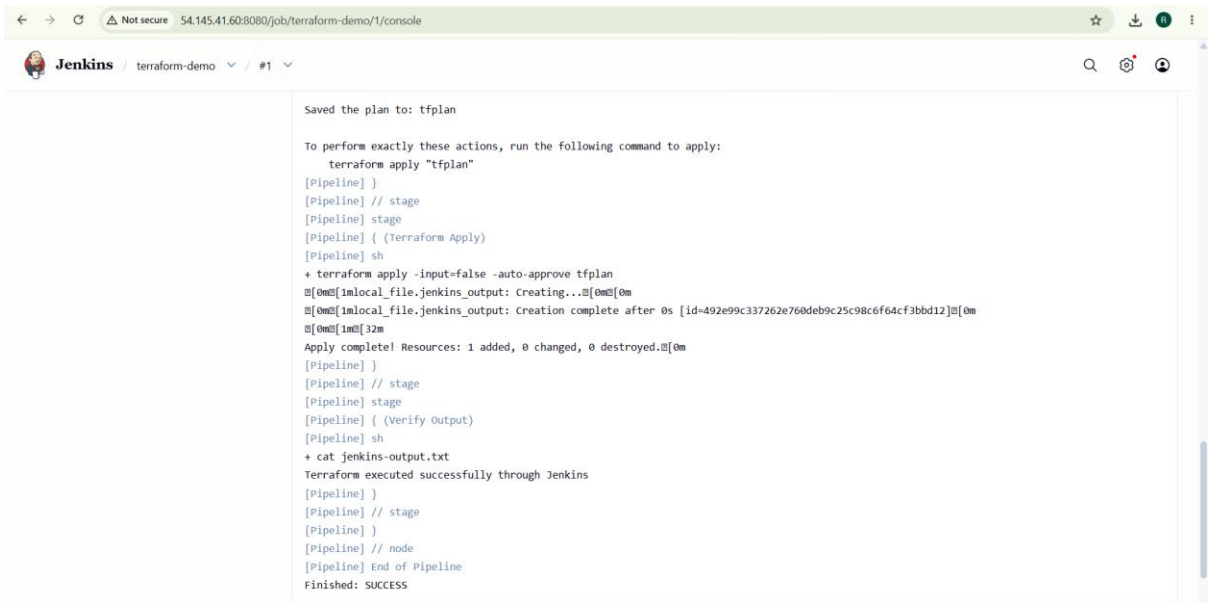


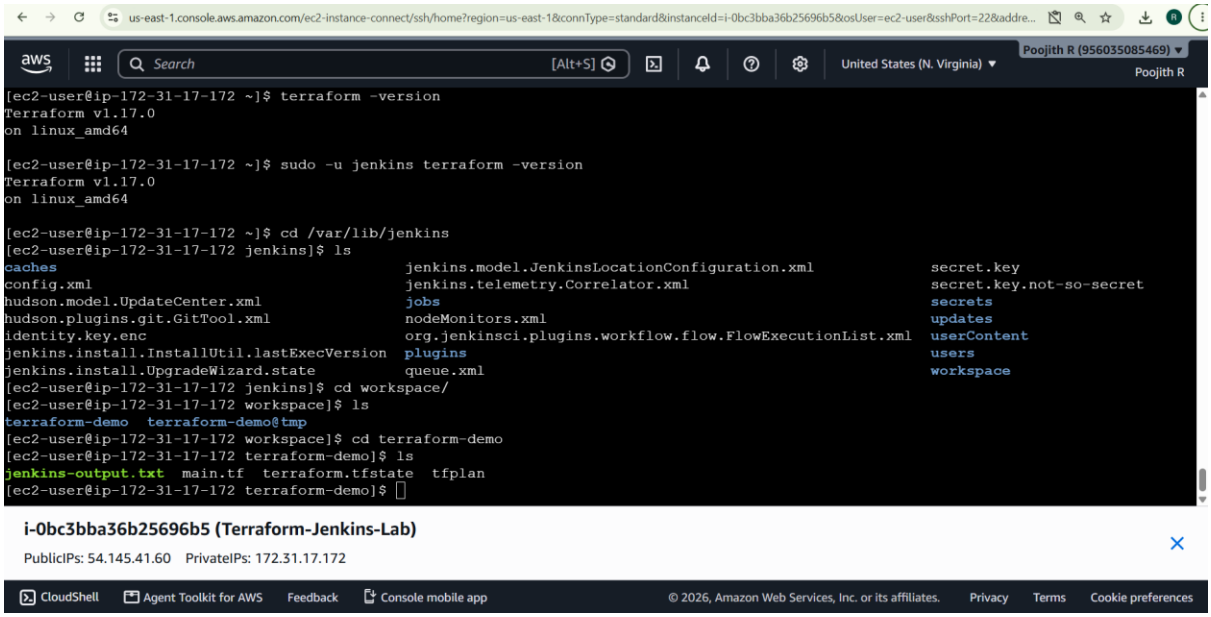

```
[ec2-user@ip-172-31-17-172 ~]$ sudo dnf install -y dnf-plugins-core
Last metadata expiration check: 0:04:02 ago on Mon Oct 26 12:26:35 2026.
Package dnf-plugins-core-4.3.0-13.amzn2023.0.7.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-17-172 ~]$ sudo dnf config-manager --add-repo https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
Adding repo from: https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
[ec2-user@ip-172-31-17-172 ~]$ sudo dnf install -y terraform
Hashicorp Stable - x86_64                                                    42 MB/s | 1.9 MB     00:00
Last metadata expiration check: 0:00:01 ago on Mon Oct 26 12:30:49 2026.
Dependencies resolved.
================================================================================================================================
 Package                    Architecture          Version                                  Repository                    Size
================================================================================================================================
Installing:
 terraform                  x86_64                1.17.0-1                                 hashicorp                     37 M
Installing dependencies:
 git                        x86_64                2.50.1-1.amzn2023.0.1                    amazonlinux                   53 k
 git-core                   x86_64                2.50.1-1.amzn2023.0.1                    amazonlinux                  4.9 M
 git-core-doc               noarch                2.50.1-1.amzn2023.0.1                    amazonlinux                  2.8 M
 perl-Error                 noarch                1:0.17030-2.amzn2023.0.1                 amazonlinux                   42 k
 perl-File-Find             noarch                1.37-477.amzn2023.0.11                   amazonlinux                   26 k
```

i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172
CloudShell Agent Toolkit for AWS Feedback Console mobile app © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0bc3bba36b25696b5&osUser=ec2-user&sshPort=22&addre...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
Installing : git-2.50.1-1.amzn2023.0.1.x86_64 8/9
Installing : terraform-1.17.0-1.x86_64 9/9
Running scriptlet: terraform-1.17.0-1.x86_64 9/9
Verifying : git-2.50.1-1.amzn2023.0.1.x86_64 1/9
Verifying : git-core-2.50.1-1.amzn2023.0.1.x86_64 2/9
Verifying : git-core-doc-2.50.1-1.amzn2023.0.1.noarch 3/9
Verifying : perl-Error-1:0.17030-2.amzn2023.0.1.noarch 4/9
Verifying : perl-File-Find-1.37-477.amzn2023.0.11.noarch 5/9
Verifying : perl-Git-2.50.1-1.amzn2023.0.1.noarch 6/9
Verifying : perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64 7/9
Verifying : perl-lib-0.65-477.amzn2023.0.11.x86_64 8/9
Verifying : terraform-1.17.0-1.x86_64 9/9
Installed:
git-2.50.1-1.amzn2023.0.1.x86_64 git-core-2.50.1-1.amzn2023.0.1.x86_64 git-core-doc-2.50.1-1.amzn2023.0.1.noarch
perl-Error-1:0.17030-2.amzn2023.0.1.noarch perl-File-Find-1.37-477.amzn2023.0.11.noarch perl-Git-2.50.1-1.amzn2023.0.1.noarch
perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64 perl-lib-0.65-477.amzn2023.0.11.x86_64 terraform-1.17.0-1.x86_64
Complete!
[ec2-user@ip-172-31-17-172 ~]$ terraform -version
Terraform v1.17.0
on linux_amd64
[ec2-user@ip-172-31-17-172 ~]$
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172


Not secure 54.145.41.60:8080/job/terraform-demo/1/console
Jenkins / terraform-demo / #1
Status
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Poojith R
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/terraform-demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Check Terraform)
[Pipeline] sh
+ terraform -version
Terraform v1.17.0
on linux_amd64
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Create Terraform Template)
[Pipeline] writeFile
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Terraform Init)
[Pipeline] sh
+ terraform init -input=false
[0m[1mInitializing the backend...[0m


Not secure 54.145.41.60:8080/job/terraform-demo/1/console
Jenkins / terraform-demo / #1
Saved the plan to: tfplan
To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Terraform Apply)
[Pipeline] sh
+ terraform apply -input=false -auto-approve tfplan
[0m[1mlocal_file.jenkins_output: Creating...[0m[0m
[0m[1mlocal_file.jenkins_output: Creation complete after 0s [id=492e99c337262e760deb9c25c98c6f64cf3bbd12][0m
[0m[1m[32m
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.[0m
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Output)
[Pipeline] sh
+ cat jenkins-output.txt
Terraform executed successfully through Jenkins
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-0bc3bba36b25696b5&osUser=ec2-user&sshPort=22&addre...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
[ec2-user@ip-172-31-17-172 ~]$ terraform -version
Terraform v1.17.0
on linux_amd64
[ec2-user@ip-172-31-17-172 ~]$ sudo -u jenkins terraform -version
Terraform v1.17.0
on linux_amd64
[ec2-user@ip-172-31-17-172 ~]$ cd /var/lib/jenkins
[ec2-user@ip-172-31-17-172 jenkins]$ ls
caches
config.xml
hudson.model.UpdateCenter.xml
hudson.plugins.git.GitTool.xml
identity.key.enc
jenkins.install.InstallUtil.lastExecVersion
jenkins.install.UpgradeWizard.state
jenkins.model.JenkinsLocationConfiguration.xml
jenkins.telemetry.Correlator.xml
jobs
nodeMonitors.xml
org.jenkinsci.plugins.workflow.flow.FlowExecutionList.xml
plugins
queue.xml
secret.key
secret.key.not-so-secret
secrets
updates
userContent
users
workspace
[ec2-user@ip-172-31-17-172 jenkins]$ cd workspace/
[ec2-user@ip-172-31-17-172 workspace]$ ls
terraform-demo terraform-demo@tmp
[ec2-user@ip-172-31-17-172 workspace]$ cd terraform-demo
[ec2-user@ip-172-31-17-172 terraform-demo]$ ls
jenkins-output.txt main.tf terraform.tfstate tfplan
[ec2-user@ip-172-31-17-172 terraform-demo]$
i-0bc3bba36b25696b5 (Terraform-Jenkins-Lab)
PublicIPs: 54.145.41.60 PrivateIPs: 172.31.17.172
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
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Terms
Cookie preferences